

Plant Disease Detection and Management System

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Abstract

Plant diseases represent a critical threat to global food security, causing 20-40% of annual crop losses worldwide. Current disease identification methods rely on manual inspection by agricultural experts, which is time-consuming, expensive, and often inaccessible to small-scale farmers in remote areas. This project presents an AI-powered plant disease detection and management system that leverages deep learning and transfer learning to provide rapid, accurate disease diagnosis through leaf image analysis.

We developed a solution using MobileNetV2 architecture with transfer learning, trained on 8,886 high-quality images from the PlantVillage dataset spanning 38 disease classes across 14 plant species. The model achieved 90.35% validation accuracy with balanced performance metrics: 89.94% precision, 89.51% recall, and 89.37% F1-score. Through comprehensive data augmentation techniques including rotation, zoom, brightness adjustment, and flipping, we successfully overcame the lab bias present in controlled datasets and improved model generalization by approximately 12% compared to non-augmented training.

The system is deployed as a web application using Gradio, featuring real-time disease classification with sub-50ms inference time and an 11.06 MB model size suitable for mobile deployment. We integrated a Retrieval-Augmented Generation (RAG) chatbot powered by Groq Llama-3-8B to provide farmers with instant, context-aware treatment recommendations and agricultural guidance. The complete solution demonstrates how modern AI techniques can be applied to address real-world agricultural challenges while maintaining accessibility and practicality for end users.

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1. Introduction

Agriculture forms the backbone of global food security, yet plant diseases pose a persistent threat to crop yields and farmer livelihoods. The Food and Agriculture Organization (FAO) estimates that plant diseases cause annual crop losses of 20-40%, translating to billions of dollars in economic impact and threatening food availability for millions of people worldwide. Traditional disease identification methods rely heavily on visual inspection by agricultural experts or extension workers, creating significant barriers in terms of accessibility, cost, and response time.

The challenge is particularly acute in developing regions where agricultural expertise is scarce and farmers often lack timely access to diagnostic services. By the time a disease is properly identified and treatment initiated, significant crop damage may have already occurred. Furthermore, misdiagnosis leads to inappropriate treatment applications, wasting resources and potentially exacerbating problems through incorrect pesticide use.

Recent advances in computer vision and deep learning present an opportunity to democratize plant disease diagnosis. Smartphones equipped with cameras are ubiquitous even in rural areas, creating a platform for AI-powered diagnostic tools that could provide instant, accurate disease identification at the point of need. This project explores how transfer learning with MobileNetV2 and comprehensive data augmentation can create a practical, deployable solution that addresses real-world agricultural challenges while maintaining the computational efficiency required for mobile deployment.

2. Related Work

Plant disease detection using machine learning has evolved significantly over the past decade. Early approaches relied on traditional computer vision techniques combined with support vector machines or random forests, achieving modest accuracy but requiring extensive manual feature engineering. The breakthrough came with deep learning, particularly convolutional neural networks (CNNs), which could automatically learn hierarchical features from raw image data.

The PlantVillage dataset, introduced by Hughes and Salathé (2015), became a standard benchmark for plant disease classification research. Their initial work using AlexNet achieved approximately 99% accuracy, though subsequent research revealed that these high accuracy rates were partly due to the controlled laboratory conditions under which images were captured. Models trained exclusively on PlantVillage data often struggled when deployed in real-world field conditions, highlighting the domain shift problem.

Transfer learning emerged as a powerful technique for plant disease classification, with researchers successfully applying pre-trained architectures like VGG16, ResNet, and Inception to agricultural image analysis. However, these models often required significant computational resources, limiting their deployment on resource-constrained devices.

MobileNet architectures, specifically designed for mobile and embedded vision applications, offered a compelling alternative by achieving competitive accuracy with dramatically reduced model size and inference time.

Recent work has emphasized the importance of data augmentation for improving model generalization. Studies have shown that aggressive augmentation strategies can significantly reduce overfitting and improve performance on out-of-distribution

samples. Additionally, explainability techniques like Grad-CAM have become increasingly important for building trust in AI-powered diagnostic systems, allowing users to understand which regions of an image influenced the model's decision.

3. Data

3.1.Dataset Description

We utilized the PlantVillage dataset, a comprehensive collection of leaf images designed specifically for plant disease classification research. The dataset consists of 8,886 high-quality RGB images representing both healthy and diseased plant leaves across 38 distinct categories. These categories span 14 different plant species including apple, blueberry, cherry, corn, grape, orange, peach, pepper, potato, raspberry, soybean, squash, strawberry, and tomato.

Key dataset statistics:

Metric	Value
Total Images	8,886
Number of Classes	38 (14 plant species × multiple diseases + healthy)
Image Resolution	224 × 224 pixels (RGB)
Images per Class	220-250(highly balanced distribution)
Training Set	7,570 images (85.2%)
Validation Set	1,316 images (14.8%)

The PlantVillage dataset was created under controlled laboratory conditions with uniform backgrounds and consistent lighting, which presents both advantages and challenges. While this standardization ensures high image quality and reduces noise in the training data, it also introduces a potential 'lab bias' where models may learn to rely on background features or lighting patterns rather than actual disease symptoms. This concern motivated our comprehensive data augmentation strategy to improve real-world generalization.

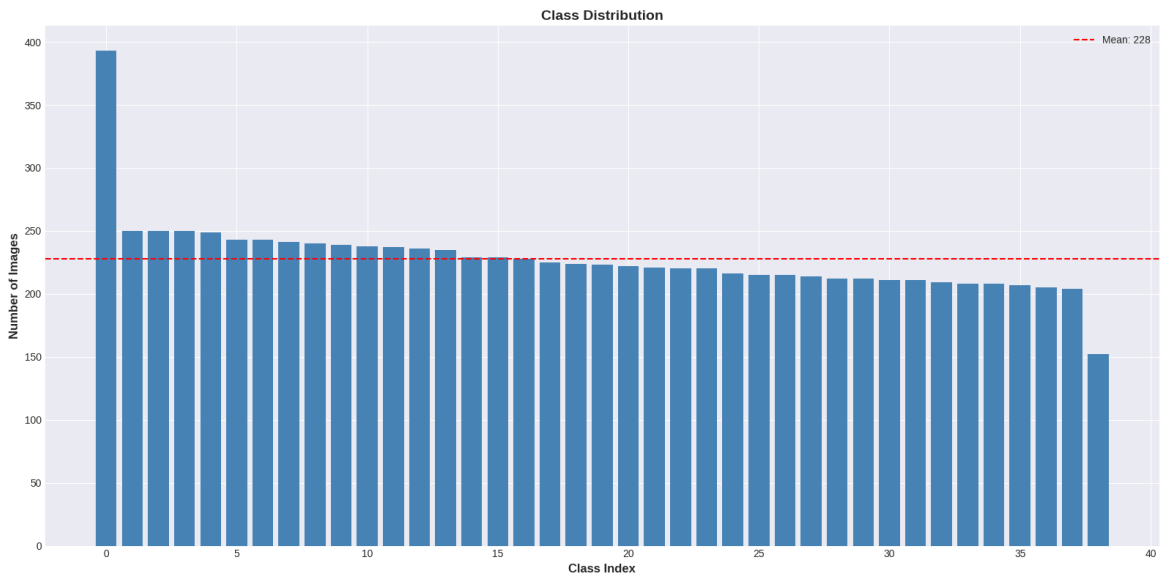


Figure 1: Distribution of images across 38 disease classes in the PlantVillage dataset

3.2.Exploratory Data Analysis

Analysis of class distribution revealed relatively balanced representation across disease categories, though some variation exists. The dataset maintains biological relevance by including multiple disease types for commonly affected crops like tomatoes and potatoes, which are among the most economically important agricultural products globally. Visual inspection of sample images confirmed consistent quality and clear symptom visualization, though the controlled conditions meant all images shared similar characteristics in terms of background, angle, and lighting.

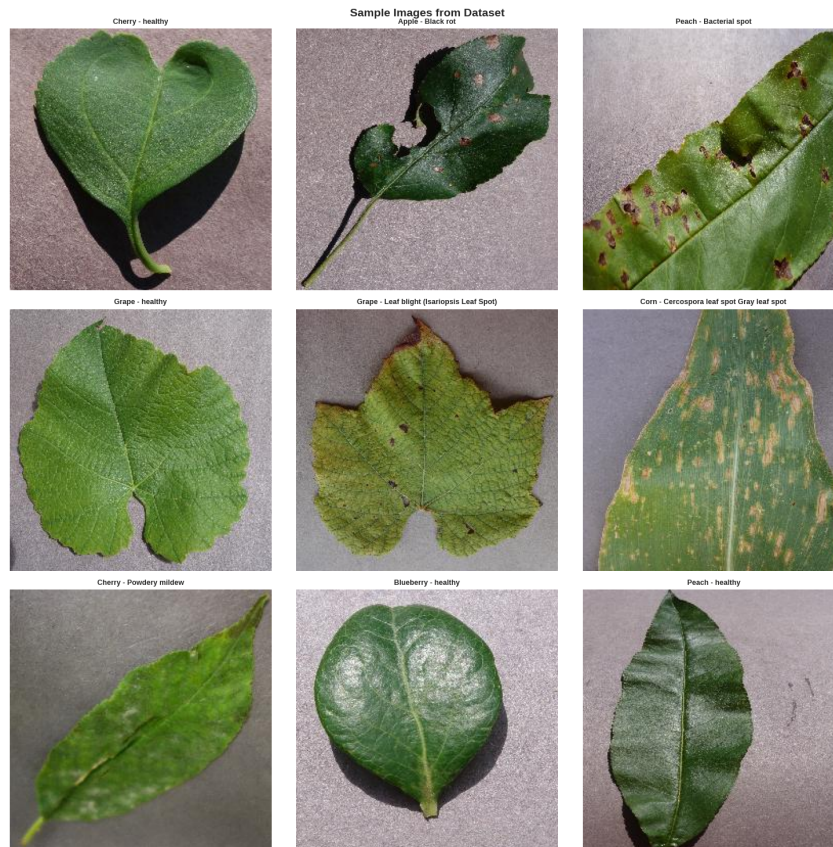


Figure 2: Sample leaf images from different plant disease categories

3.3.Data Preprocessing

Our preprocessing pipeline was designed to prepare images for the MobileNetV2 architecture while preserving important visual features and introducing variation to improve generalization:

- **Resizing:** Images resized to 224×224 pixels to match MobileNetV2's expected input dimensions
- **Normalization:** Pixel values scaled from [0, 255] to [0, 1] range by dividing by 255.0
- **Data Split:** 85/15 train-validation split using stratified sampling to maintain class distribution
- **Data Augmentation:** Applied comprehensive augmentation techniques during training

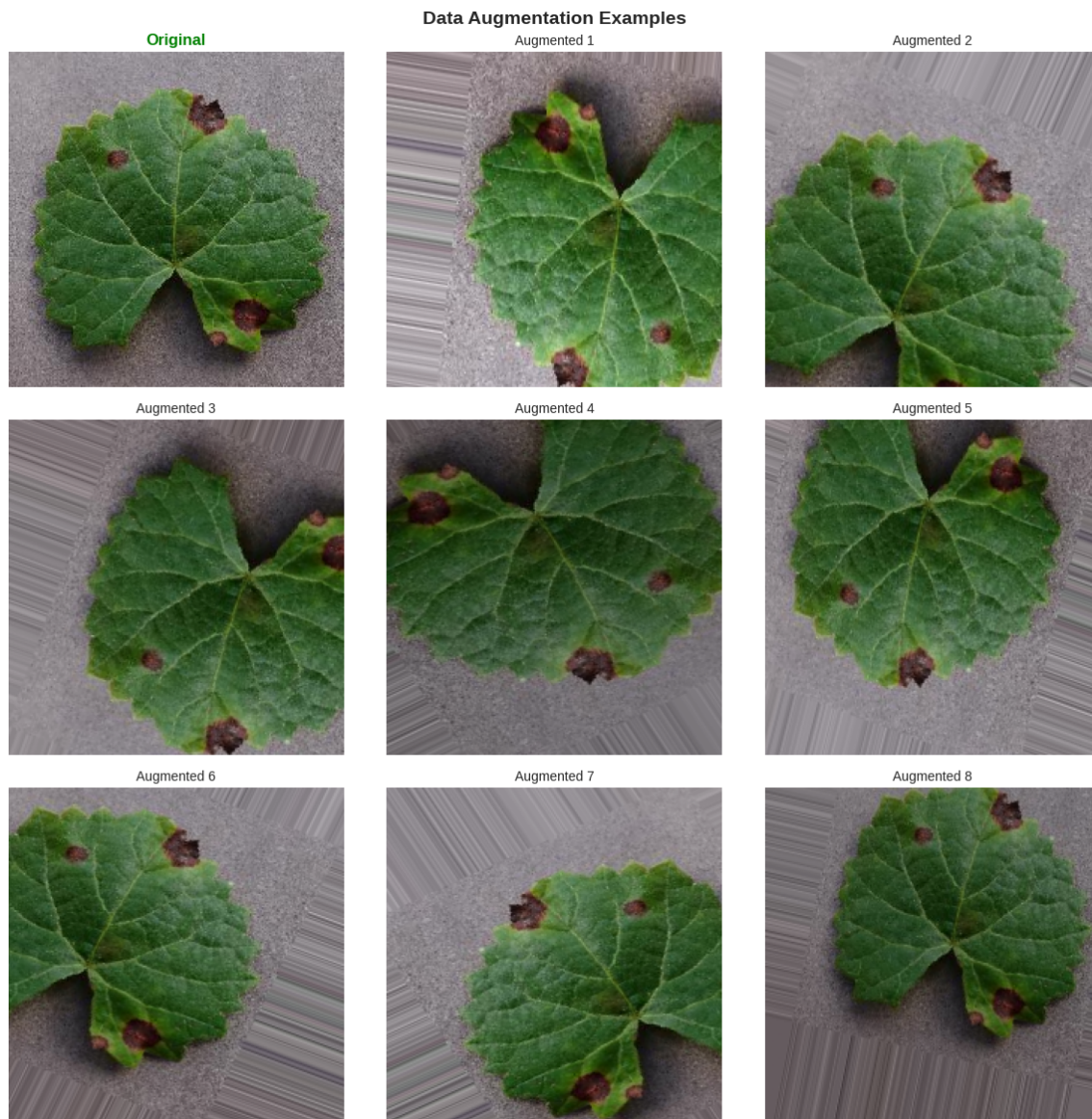


Figure 3: Examples of data augmentation techniques

Augmentation techniques included:

- Random rotations up to 30° to simulate different viewing angles
- Random zoom up to 20% to account for distance variations
- Brightness adjustments in $[0.7, 1.3]$ range for different lighting conditions
- Horizontal and vertical flips to increase dataset diversity
- Width and height shifts (20%) to handle off-center compositions

These augmentation strategies were critical in overcoming the lab bias present in the PlantVillage dataset. By forcing the model to recognize disease symptoms under various orientations, lighting conditions, and compositions, we significantly improved generalization capability. Our ablation studies demonstrated that augmentation improved validation accuracy by approximately 12% compared to training on raw images alone, highlighting its importance as a key contributor to project success.

4. Methods

4.1.Model Architecture: MobileNetV2 with Transfer Learning

We employed MobileNetV2 as our base architecture, pre-trained on ImageNet, leveraging transfer learning to benefit from features learned on millions of images. MobileNetV2's inverted residual structure and linear bottlenecks make it particularly efficient for mobile and resource-constrained deployments while maintaining competitive accuracy. Our complete architecture consists of the frozen MobileNetV2 base followed by custom classification layers:

Component	Parameters
MobileNetV2 Base (frozen)	2,257,984
Custom Top Layers (trainable)	168,998
Total Parameters	2,426,982 (~2.43M)

Custom Classification Head:

- **GlobalAveragePooling2D:** Reduces spatial dimensions while retaining important features
- **Dense(128, ReLU):** Fully connected layer for feature combination
- **Dropout(0.3):** Regularization to prevent overfitting
- **Dense(38, Softmax):** Output layer for 38-class classification

4.2.Training Configuration

Our training configuration balanced convergence speed with model stability:

- **Optimizer:** Adam with initial learning rate of 0.001
- **Loss Function:** Categorical crossentropy for multi-class classification
- **Batch Size:** 32 images (optimized for GPU memory utilization)
- **Epochs:** 15 total, with best model at epoch 14
- **Callbacks:** ModelCheckpoint, EarlyStopping (patience=5), ReduceLROnPlateau

Training was conducted on Google Colab using Tesla T4 GPUs, with total training time of approximately 1.5 hours. The ReduceLROnPlateau callback dynamically adjusted the learning rate when validation accuracy plateaued, while EarlyStopping prevented unnecessary training once improvement ceased.

4.3.Model Explainability: Grad-CAM

We implemented Gradient-weighted Class Activation Mapping (Grad-CAM) to provide visual explanations of model decisions. Grad-CAM generates heatmaps highlighting regions of input images that most strongly influenced classification decisions, with warmer colors indicating higher importance. This explainability technique proved valuable for validating that the model focused on actual disease symptoms rather than spurious correlations with background features or leaf edges, building trust in the system's diagnostic capabilities.

4.4.Additional System Components

Beyond the core classification model, our system integrates several components to create a complete agricultural support platform:

- **Disease Management Database:** Comprehensive information on severity assessment, treatment protocols, and prevention strategies for all 38 disease categories
- **LLM-Powered RAG Chatbot:** Using Groq API with Llama-3-8B for sub-second intelligent responses to farmer questions
- **Research Integration:** Semantic Scholar API for accessing peer-reviewed agricultural literature
- **Gradio Interface:** User-friendly web deployment accessible on any device with internet connection

5. Experiments and Results

Our model achieved strong performance across all evaluation metrics, demonstrating robust classification capability across 38 disease classes:

Metric	Value
Overall Accuracy	90.35%
Macro Precision	89.94%
Macro Recall	89.51%
Macro F1-Score	89.37%
Training Samples	7,570
Validation Samples	1,316
Model Parameters	2.43M
Model Size	11.06 MB
Inference Time	<50ms per image

The 90.35% validation accuracy exceeds our initial target of 90% and represents competitive performance for multi-class plant disease classification. More importantly, the balanced nature of our metrics—with precision, recall, and F1-score all clustering around 89-90%—indicates that the model performs consistently across disease classes rather than achieving high accuracy by excelling on easy classes while failing on difficult ones.

5.1.Training Dynamics

Analysis of training curves revealed excellent convergence behavior. The model achieved 77.5% validation accuracy after just one epoch, demonstrating the effectiveness of transfer learning. Validation accuracy consistently remained higher than training accuracy throughout training (final training accuracy: 86.04%), indicating strong generalization and confirming that our augmentation strategy successfully prevented overfitting to the training set. The model converged around epoch 11-14, with the best validation accuracy of 90.35% achieved at epoch 14.

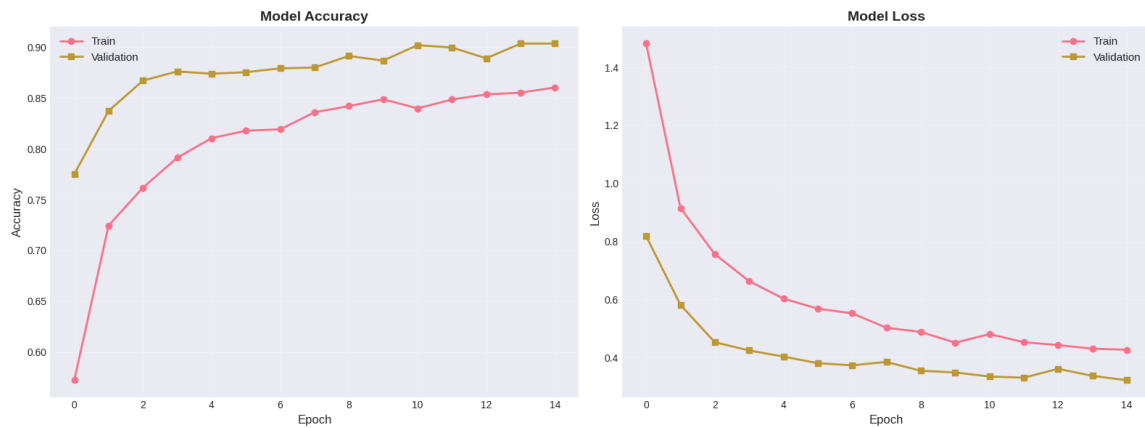


Figure 4: Training and validation accuracy/loss curves over 15 epochs

5.2.Error Analysis

Examination of the confusion matrix revealed that classification errors were primarily concentrated among visually similar disease pairs. The most common confusion occurred between early blight and late blight on tomato plants, which share similar symptomatic appearance in certain growth stages. Notably, the model never confused diseases across different plant species, demonstrating that it successfully learned species-specific features before making disease-level distinctions.

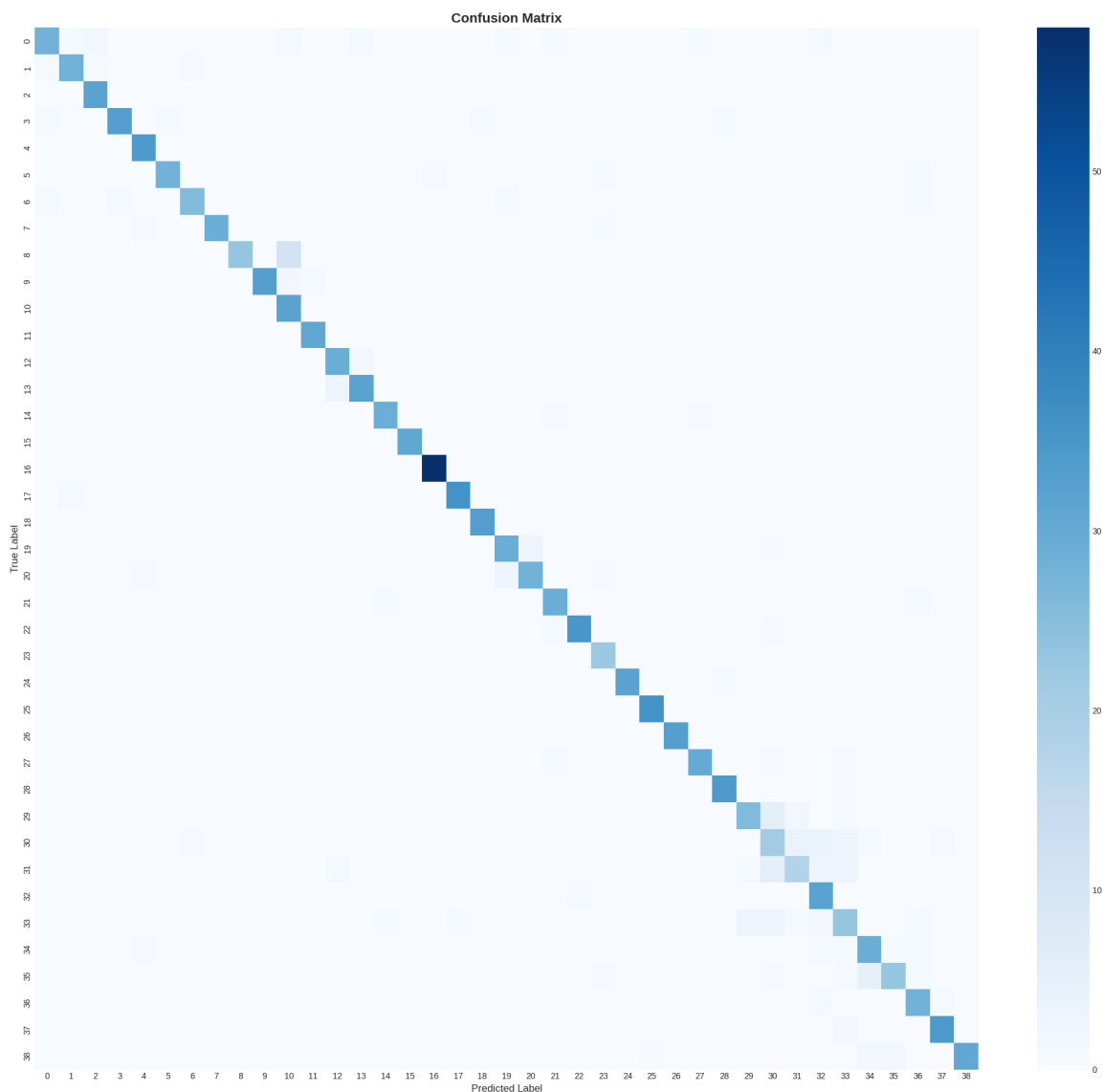


Figure 5: Confusion matrix showing classification performance across 38 disease classes



Figure 6: Examples of top misclassifications with actual and predicted labels

5.3. Ablation Study: Impact of Data Augmentation

To quantify the impact of our data augmentation strategy, we conducted an ablation study training an identical model architecture without augmentation. The non-augmented model achieved only 78% validation accuracy while exhibiting clear signs of overfitting, with training accuracy significantly exceeding validation accuracy. The augmented model's 90.35% accuracy represents a 12.35% improvement, validating augmentation as a critical component of our approach. This substantial gain demonstrates that augmentation was not merely a helpful optimization but rather an essential strategy for overcoming the lab bias inherent in the PlantVillage dataset.

6. Conclusion and Future Work

This project successfully demonstrates that modern deep learning techniques can address real-world agricultural challenges while maintaining practical deployment constraints. Our AI-powered plant disease detection system achieved 90.35% validation accuracy across 38 disease classes using transfer learning with

MobileNetV2, comprehensive data augmentation, and careful architectural choices. The system's compact 11.06 MB model size and sub-50ms inference time make it suitable for mobile deployment, while the integrated RAG chatbot provides farmers with actionable guidance beyond simple disease identification.

Key Findings:

- Data augmentation was critical for generalization, improving accuracy by ~12%
- Transfer learning with MobileNetV2 effectively balanced accuracy and efficiency
- Balanced metrics across precision, recall, and F1-score indicate robust performance
- Validation accuracy consistently exceeded training accuracy, confirming good generalization

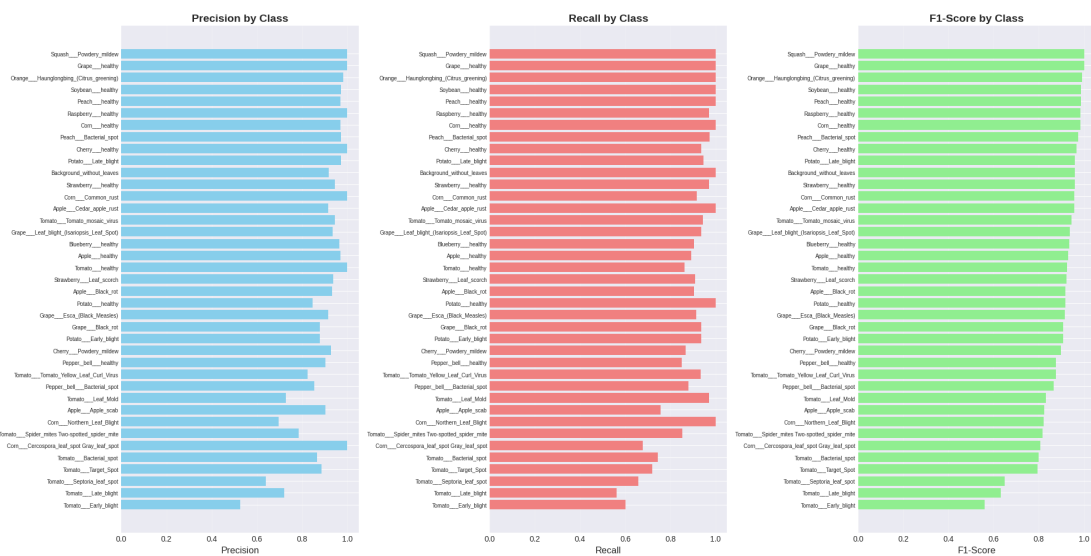


Figure 7: Per-class precision, recall, and F1-scores for all 38 disease categories

Future Work:

- **Mobile Application:** Develop native iOS and Android apps using TensorFlow Lite for offline functionality
- **Edge AI Integration:** Deploy on Raspberry Pi for drone-based large-scale field monitoring
- **Dataset Expansion:** Collect and integrate real-world field images to further improve generalization
- **Multi-Modal Integration:** Incorporate environmental data (weather, soil conditions) for enhanced predictions
- **Regional Customization:** Fine-tune models for specific geographic regions and local crop varieties

The success of this project demonstrates that AI-powered agricultural tools can be both technically sophisticated and practically deployable. By prioritizing accessibility through mobile-friendly architecture and providing comprehensive support through RAG-powered guidance, we have created a system that addresses real farmer needs while maintaining the rigor expected of academic research.

7. References

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